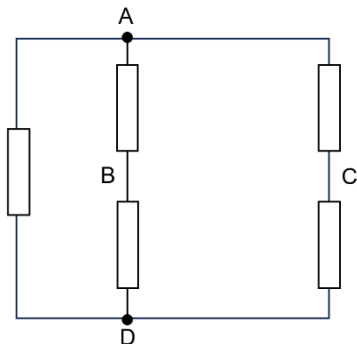


distance l , the cross sectional area decreases, resulting in larger increases in R .

23

C

Since potentials at points C and B are equal, we can remove the resistor between them.



$$\frac{1}{R_{AD}} = \frac{1}{6.0} + \frac{1}{12.0} + \frac{1}{12.0}$$

$$\frac{1}{R_{eff}} = \frac{4}{12.0}$$

$$R_{eff} = 3.0 \, \Omega$$

Note that this question has been simplified. The original question has the setup below:

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C2

24

D

The forces acting on the vertical sides of the coil $= nBl$ since the current in the vertical